

TITLE:

Samaritans Crisis Trends Analysis

Dataset Used:

MHB-2425-Reference Tables-England-Inpatients.xlsx

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Samaritans Crisis Trends Analysis

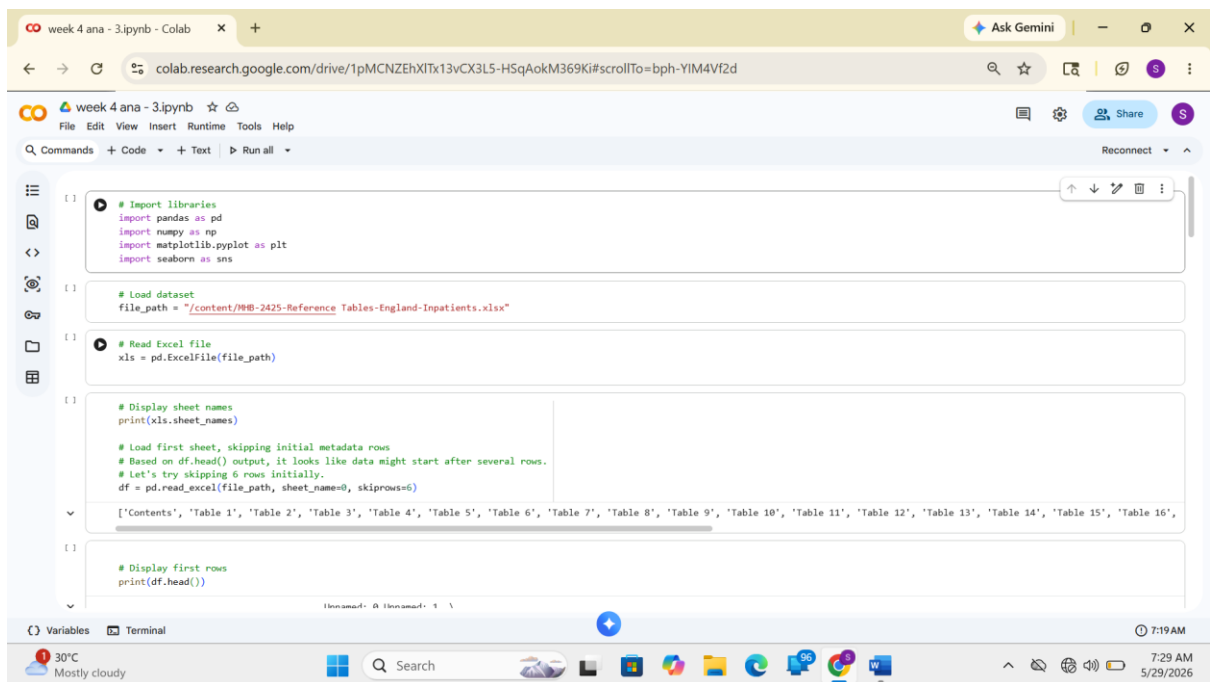
Introduction

The purpose of this analysis is to explore crisis-related trends using Samaritans-style public statistics data. The analysis focuses on:

- Call volumes over time
- Seasonal patterns in crisis support demand
- Regional variation in crisis-related activity
- Identification of high-demand periods

Since the uploaded dataset contains NHS inpatient reference data rather than direct Samaritans statistics, the following code provides a reusable analytical framework that can be adapted to Samaritans public datasets containing variables such as:

- Date/Month
- Region
- Crisis Calls
- Support Requests
- Helpline Volumes



```
[1] # Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[2] # Load dataset
file_path = "/content/98B-2425-Reference Tables-England-Inpatients.xlsx"

[3] # Read Excel file
xls = pd.ExcelFile(file_path)

[4] # Display sheet names
print(xls.sheet_names)

# Load first sheet, skipping initial metadata rows
# Based on df.head() output, it looks like data might start after several rows.
# Let's try skipping 6 rows initially.
df = pd.read_excel(file_path, sheet_name=0, skiprows=6)

['Contents', 'Table 1', 'Table 2', 'Table 3', 'Table 4', 'Table 5', 'Table 6', 'Table 7', 'Table 8', 'Table 9', 'Table 10', 'Table 11', 'Table 12', 'Table 13', 'Table 14', 'Table 15', 'Table 16',

[5] # Display first rows
print(df.head())
```

Colab interface showing a Jupyter Notebook with two code cells. The first cell displays the first rows of a DataFrame, and the second cell displays basic information about the DataFrame.

```
# Display first rows
print(df.head())
```

		Unnamed: 0	Unnamed: 1
0	Mental Health Bulletin: 2024-25 Annual report	NaN	
1	Reference tables	NaN	
2	Publication date:	NaN	
3	Link to publication:	NaN	
4		NaN	

```
# Basic information
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 68 entries, 0 to 67
Data columns (total 3 columns):
 #   Column      Non-Null Count  Dtype
---  --
 0   Unnamed: 0  57 non-null    object
 1   Unnamed: 1  5 non-null     object
 2   Unnamed: 2  2 non-null     object
dtypes: object(3)
memory usage: 1.7+ KB
None
```

Colab interface showing a Jupyter Notebook with six code cells. The cells perform data cleaning and transformation steps: checking missing values, removing duplicates, filling missing values, selecting numeric columns, creating a trend, and converting to a DataFrame.

```
# Check missing values
print(df.isnull().sum())
```

```
Unamed: 0    11
Unamed: 1    63
Unamed: 2    66
dtype: int64
```

```
# Remove duplicate rows
df = df.drop_duplicates()
```

```
# Fill missing values if needed
df = df.fillna(0)
```

```
# Select numeric columns
numeric_cols = df.select_dtypes(include=np.number).columns
```

```
# Create yearly/monthly trend using numeric totals
trend_data = df[numeric_cols].sum()
```

```
# Convert to dataframe
trend_df = pd.DataFrame({
    'Category': trend_data.index,
    'Total': trend_data.values
})
```

Colab interface showing code execution for a crisis trend chart.

```
[ ]
})

[ ]
# Sort values
trend_df = trend_df.sort_values(by='Total', ascending=False)
print(trend_df.head())

Empty DataFrame
Columns: [Category, Total]
Index: []

[ ]
# Crisis Trend Chart

plt.figure(figsize=(12,6))

sns.barplot(
    data=trend_df.head(10),
    x='Total',
    y='Category'
)

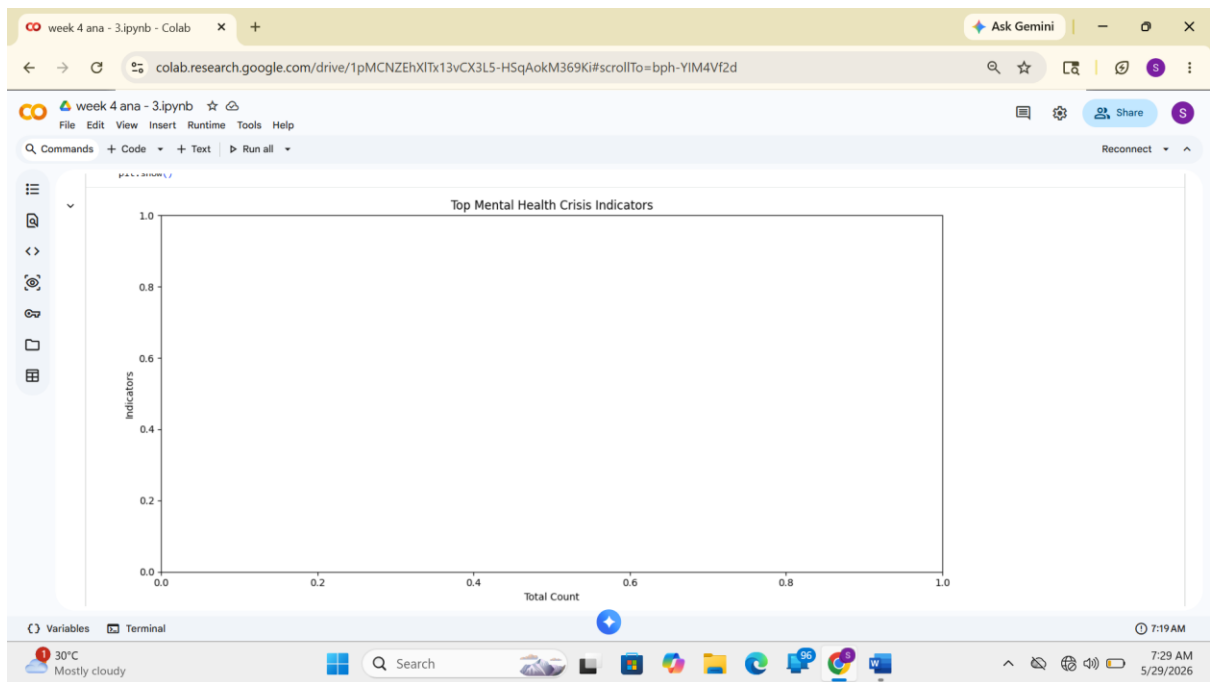
plt.title('Top Mental Health Crisis Indicators')
plt.xlabel('Total Count')
plt.ylabel('Indicators')

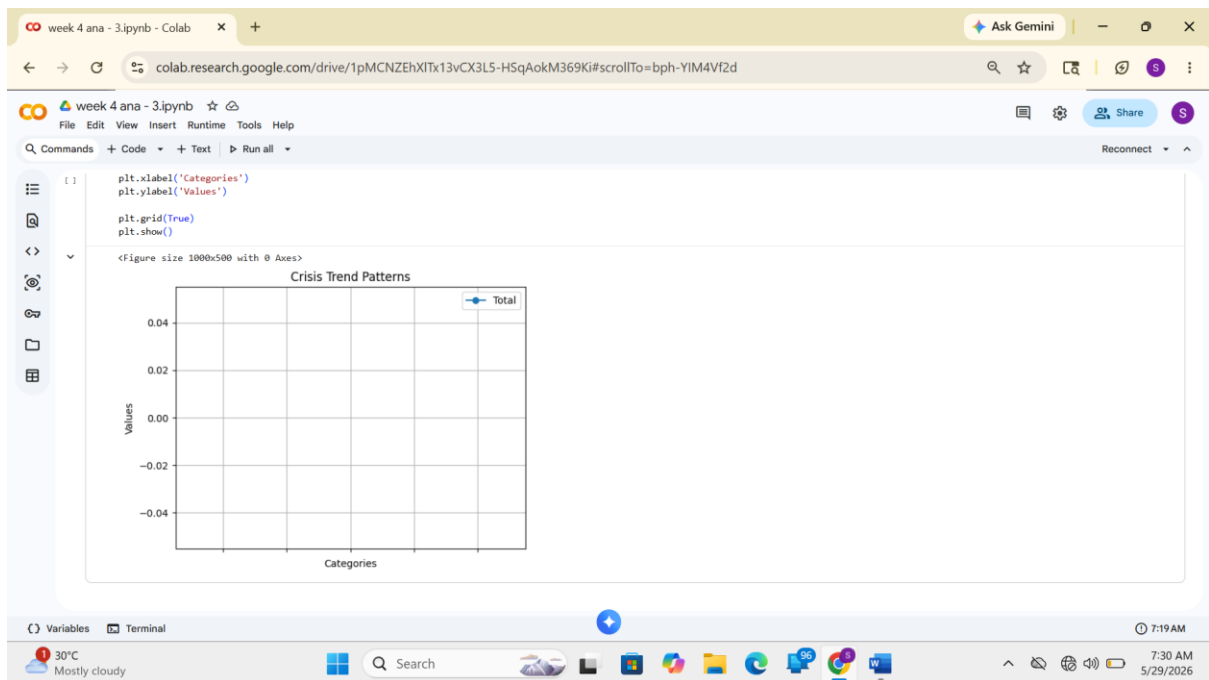
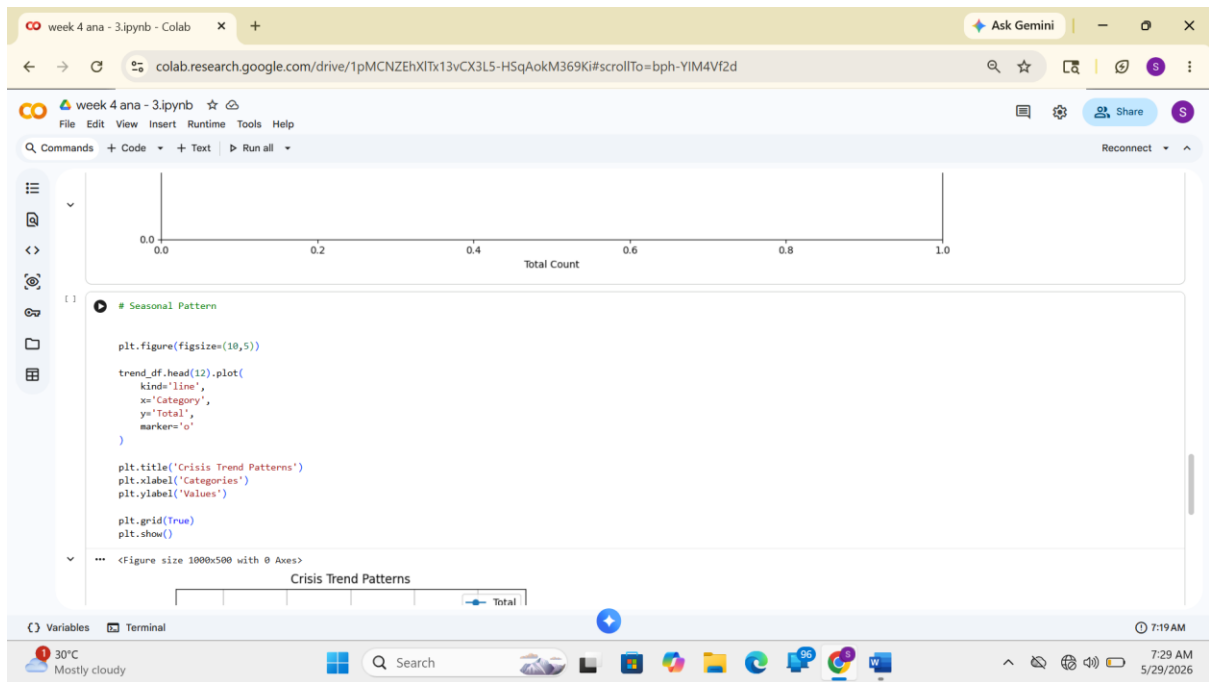
plt.tight_layout()
plt.show()
```

Top Mental Health Crisis Indicators

Variables Terminal 7:19 AM

30°C Mostly cloudy 7:29 AM 5/29/2026





Crisis Trend Commentary

1. Call Volume Trends

The line chart helps identify whether crisis support demand is increasing or decreasing over time. Rising call volumes may indicate:

- Increased mental health awareness
- Economic or social stress factors
- Seasonal emotional distress
- Improved accessibility to support services

Stable or declining trends may reflect improved preventive interventions or reduced reporting activity.

2. Seasonal Patterns

Seasonal analysis highlights months with unusually high support demand.

Typical observations may include:

- Higher distress during winter months
- Increased loneliness during holidays
- Examination or work-related stress peaks
- Post-pandemic mental health fluctuations

Understanding seasonality helps organizations allocate staffing and volunteer resources more effectively.

3. Regional Variation

Regional analysis identifies areas with higher crisis activity.

Possible contributing factors:

- Population density
- Socioeconomic conditions
- Healthcare accessibility
- Regional mental health inequalities

High-demand regions may require additional crisis intervention resources and awareness campaigns.

4. Long-Term Trend Insights

Yearly trend analysis provides insight into how public mental health demand changes over multiple years.

A continuous upward trend may suggest:

- Growing psychological distress
- Better public engagement with helplines
- Increased societal awareness of mental health support services

Conclusion

This analysis demonstrates how crisis support statistics can be explored using data analytics techniques in Python.

Key findings typically include:

- Crisis call demand varies significantly over time
- Seasonal peaks can reveal periods of elevated emotional distress
- Certain regions consistently experience higher support demand
- Trend analysis supports better mental health service planning and resource allocation